

## Homework 1

Big Data (BAX 423)

Instructor: Rahul Makhijani

Due: April 10, 2026 - 11:59 pm

On Canvas you will submit the following files:

- A write-up of your responses to Part 1.
- iPython notebooks with all cells ran for Part 2 and 3 (answer any questions in the notebook).
- Python code and hmtl file you generated in Part 4.
- A short video (.mp4) of you describing what your group did in Part 2.

### Part 1 - ML Design

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Write short responses to the following questions.

1. You're building a song recommendation tool. List why and why not you might want to pick a "frontier model", a proprietary model that is the most advanced, versus an open source model.
2. Now assume you're building the model for Spotify and you have access to users' listening histories. What are some features you might want to compute? When you move from training your model to serving will each of these features require streaming data?
3. How does the requirements of your system change if you're sending users a list of recommendations that updates once a day/week (a "For you" playlist, essentially) versus if you are recommending a new song related to each song a user is currently listening to?
4. Some artists become incredibly popular suddenly. New subgenres can emerge. What can you do to minimize the **drift** of your model?

### Part 2 - Fine-tuning

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In this section we will be using the Newswire dataset of articles scanned from historical newspapers constructed by Melissa Dell and company. More info on this dataset can be found at this [link](#). Your model will be parsing historical newspaper articles from the 1960s and classifying whether or not they pertain to civil rights.

- a. Download the file `finetune.py` from Canvas.
- b. Look for all the places labeled `TODO` in the code. Do the tasks described in these comments. Note that in places where lines of code are incomplete, the original code shouldn't have to be edited.
- c. For the fine-tuning exercise, iterate to improve your results until your model reaches 93% accuracy.
- d. Discuss any modeling or data decisions you made and why they improved your results.
- e. Also keep a record of your model's peak CPU usage, GPU usage, and the final perplexity score you achieve.
- f. Submit a notebook file containing your code and the other requirements as well as a *short* screen-recording of one member of your group explaining what you did (this should be about 3 minutes or less). The easiest way for you to do this is to screen record on zoom (record locally) and submit the mp4 file generated when you end the zoom meeting.

### Part 3 - Attention Mechanism

The goal of this section is to code your own multiheaded attention with multiple types of attention masks, contrasting the performance between *\*dense\** and *\*sparse\** attention.

- a. Download the file `AttentionMechanisms.ipynb` from Canvas.
- b. Complete all `TODOs`.
- c. Answer any intuition questions with a couple sentences in the notebook.
- d. Submit your notebook with the output of all cells visible.

### Part 4 - Using an Agent with Cursor

Using an agent in Cursor, do the following:

- a. From NOAA's "Storm Events Database" get data for all weather events of a tornado rated EF2 or stronger from the years 2020 to 2025. The agent should download the data for you. (Hint: You might want to specify exactly how the agent goes about this.)
- b. Then, have the agent analyze of the states in which these tornadoes are the most common? What is the frequency of these occurrences by month? For a given state what month has the highest frequency? (You will have to tell the agent how to handle ties.)
  - Do you notice any regional pattern in the last result?

- c. Also, have the agent tabulate how many deaths there have been from these incidents. Of the fatal reports, what percentage of them mention “mobile homes” or “trailers”.
- d. Have cursor generate a pdf report of these results for you **with your group members names on it**.
- e. Have an agent, send an email to Chris (email on the syllabus) with the subject line “BAX423 Tornado Watch”. In the body of the email, tell me to “Avoid [STATES] in the month of May.” where [STATES] are the states where May has the highest frequency of tornadoes for that state. Sign off with ”From [YOUR GROUP NAME]”. Use the group name that appears on canvas.
  - **Note: You should make a throwaway Gmail account to send this from rather than sending this from your school email. One for your group.**